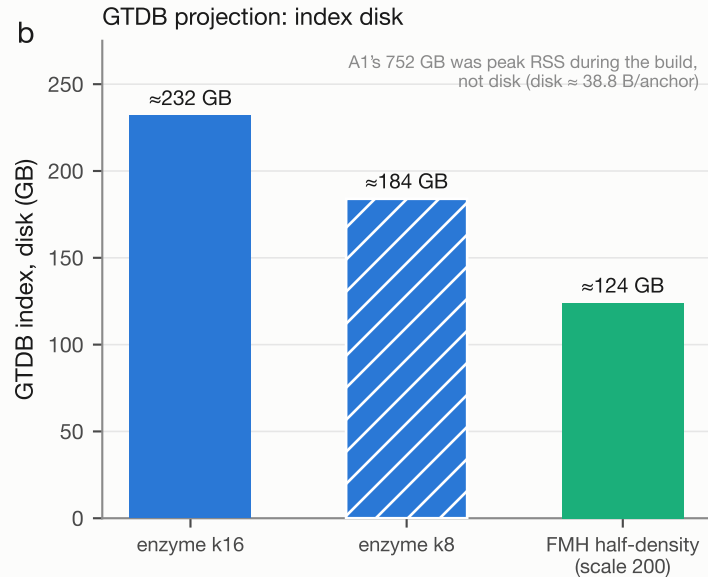
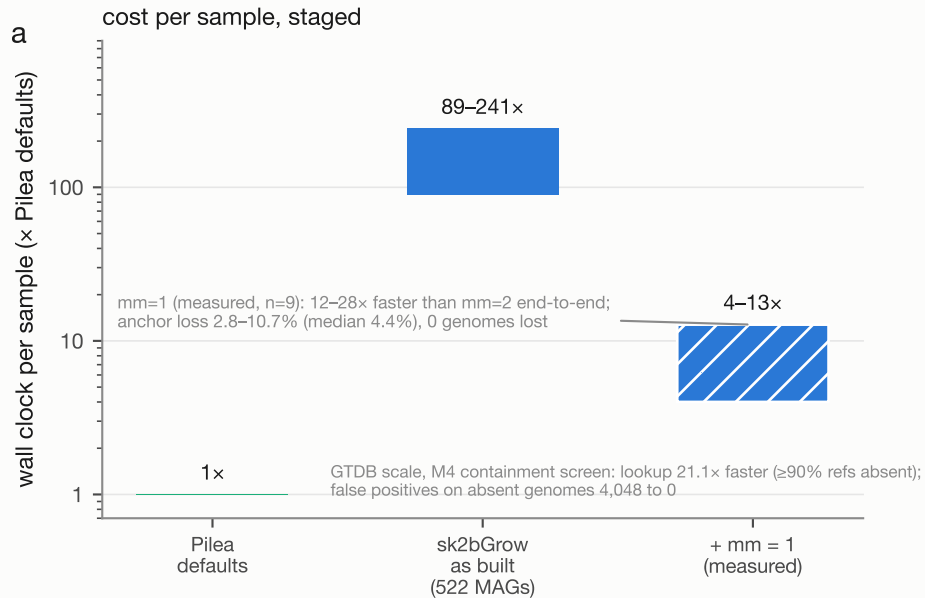




Cost narrative for scaling past single-genome benchmarks. a, wall clock per sample relative to Pileup at its defaults: on the 522-MAG RBC dataset the as-built matcher costs ≈ 90 – $240\times$ (matching reads against 24.1M anchors is 89.6% of count time); dropping to `--max-mismatch 1` (two ~ 16 -bp seeds instead of three ~ 11 -bp seeds) shortens posting lists enough to cut this to ≈ 4 – $13\times$ measured end-to-end on all nine samples (12–28× over the mm = 2 baseline), at the price of 2.8–10.7% of anchors lost (median 4.4%; 0 genomes lost; unbiased on shared anchors). At GTDB scale ($\geq 90\%$ of references absent from a sample) the M4 containment screen adds a 21.1× lookup speedup and drives false positives on absent genomes from 4,048/4,700 to 0. b, GTDB projection (136,646 species representatives ≈ 629 Gbp, 38.8 B/anchor on disk): enzyme k16 ≈ 232 GB, k8 ≈ 184 GB, half-density FMH (scale 200) ≈ 124 GB with ≤ 0.007 r lost at $\geq 1\times$. The often-quoted 752 GB is peak resident memory during a build, not disk.